

Early-Access Health Scoring: Predicting Post-Launch Retention and Community Climate from Public Steam Data

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Abstract

Early access lets a studio sell an unfinished game and develop it alongside the players who bought it. A small multiplayer studio has no practical way to judge, before full release, whether the community forming around its game will still be there afterwards. The one signal it reads is the percentage of positive reviews on its store page, which measures approval and says nothing about whether players kept playing. We test whether the public record left during early access is enough on its own to build a legible health score anticipating post-launch retention and community climate, and whether such a score improves on the free signal a studio already has. Review histories for 1,793 Steam games (3,504,533 reviews) were collected from public endpoints and reduced to game-level measures. The central measure is a retention proxy taken from the gap between a reviewer's lifetime playtime and their playtime when writing, which makes a behavioural retention quantity available without operator telemetry. Models are evaluated on a temporal split, with every scaler, weight and cutoff fitted on training games only. Community climate measured from early-access review *text* anticipated post-launch retention measured from a playtime *counter* at AUC 0.7489 against a store-page baseline of 0.6019, a gain of +0.1482 with a 95% bootstrap interval of [+0.0559, +0.2428], surviving controls for elapsed accrual time, game size and the baseline itself. Score bands committed in advance, never shown the abandoned games, still separated the 424 titles that died during early access from the 545 that launched by 8.8 points. The score does not beat the free baseline at predicting satisfaction, and that negative result is reported as it stands. We also show that several stronger-looking results in this area are artefacts of measuring predictor and outcome with the same instrument: four persistence features reproduce an entire seventeen-feature retention model, because lifetime playtime is a collection-date snapshot that already contains post-launch play.

CCS Concepts

• **Information systems** → **Data mining**; • **Applied computing** → **Computer games**; • **Computing methodologies** → *Machine learning*; • **Human-centered computing** → *Empirical studies in collaborative and social computing*.

Keywords

Early access, Steam reviews, player retention, community climate, toxicity, composite indicators, temporal validation, measurement leakage

1 Introduction

Selling an unfinished game to the people who will help finish it is now an ordinary commercial strategy rather than an experiment.

Valve's early-access programme lets a studio take revenue and feedback at the same time and sets no limit on how long a title may sit in that state [17]. Lin et al. [11] characterised the practice at platform scale, establishing early access as a subject for empirical software-engineering research rather than a curiosity. For a small studio building a multiplayer game, the arrangement creates a decision it is badly equipped to make. At some point during early access it has to judge whether the community gathering around the game is one that will sustain a population after launch, or one that will not.

The signals that judgement needs are visible in principle. Players leave reviews, the reviews carry timestamps and playtime, and the text records the tone of the people writing it. In practice a studio with no analytics function reads one number, the percentage of positive reviews on its store page. That number measures how much players liked the game. It does not measure whether they kept playing it, and it does not measure how they spoke to each other while they did. The studios least able to absorb a failed launch are therefore the ones with the least warning that one is coming.

This paper asks whether the public record a game leaves during early access is enough, on its own, to build a legible health score that anticipates post-launch player retention and community climate. The question that turns out to separate good answers from flattering ones is whether the predictor and the outcome are read off the same instrument. Where they are, a high score measures persistence of a trait rather than a forecast, and Section 6.2 shows how much of the apparent performance in this area that accounts for.

We make four contributions.

A retention proxy computed entirely from public data. Every Steam review records both the author's lifetime playtime and their playtime at the moment they wrote, so the difference between the two measures how much longer that player carried on after reviewing. Aggregated over a game's reviewers it says whether a community kept playing, and it needs no access to operator telemetry. That matters because the retention literature is built almost entirely on telemetry a small studio does not have.

A cross-instrument forecast. Early-access review text predicts post-launch playtime retention at 0.7489 AUC against a free store-page baseline of 0.6019. Because text goes in and a playtime counter comes out, this is a forecast rather than an association, and it is the principal result.

A leakage diagnostic with a measured cost. Four persistence features reproduce a whole seventeen-feature retention model, and early-access and post-launch persistence correlate at a Spearman ρ of 0.867. We report the high figures, label them as associations, and keep the ablation that explains them beside them throughout.

Table 1: The closest prior work, positioned on data access, outcome, and whether predictor and outcome share a measurement instrument.

Study	Data	Outcome predicted	Small studio?	Instr. separated?
Lin et al. [11]	public Steam	none, descriptive	yes	n/a
Guzsvinecz [5]	public Steam	review share, playtime	yes	no
Tong et al. [16]	public reviews	developer topics	yes	n/a
Sun et al. [15]	public Steam	concurrent activity	yes	partly
Abdul-Rahman et al. [1]	public reviews	in-product churn label	no	partly
Morrier et al. [13]	match logs	short-term engagement	no	yes
This work	public Steam	post-launch retention	yes	yes

A survivorship comparison the literature omits. Games that never left early access are absent from every published sample, and they are the cohort a studio most wants to avoid joining. Scored by cutoffs fitted without ever seeing them, they land 8.8 points lower than games that launched.

2 Related Work

Four literatures touch this problem and none of them meets it. Work on early access describes the release model but stops short of forecasting what happens after it. Work on Steam review text has become good at extracting what players think and has largely been pointed at approval and popularity rather than at whether players stay. Work on retention and churn is precise, and it is built on telemetry that belongs to the operator. Work on toxicity in game communities has established that hostile climate carries a real cost, and has established it using data a small studio will never see. Table 1 lays the closest studies side by side on the three axes that matter here. The gap sits where all four fail at once.

2.1 Early Access

Early access is unusual among release strategies in that the platform imposes almost no structure on it. Valve requires a playable build and honest store-page information, warns buyers when a title has gone twelve months without an update, and otherwise permits a game to stay in early access indefinitely [17]. That permissiveness is what makes the period analytically interesting: the decision to leave it is the studio’s own, and the evidence for making it accumulates publicly while it is being made. Early access entered the empirical software-engineering literature through Lin et al. [11]. What that line of work established is descriptive. It characterises how the model is used, not which uses of it end well.

2.2 Public Review Analytics

Steam reviews are attractive data because they are public, time-stamped, and attached to a verified playtime figure, and the analytical literature has taken advantage of that. Topic modelling has been used to surface the concerns developers should be reading

and to compare them against what developers believe players care about [16], sentiment has been folded into models of platform visibility and player activity [15], and review-derived measures have been related to design and mechanics within a genre [5].

The ceiling on this work is the outcome it chooses. Popularity, revenue, visibility and rating all measure how a game is received, and all are available at roughly the moment the predictor is. A studio deciding whether to leave early access is not asking how well the game is received now. It is asking whether the population will still be playing in six months, which is a different quantity, measured later and by a different instrument. Section 6.2 shows what happens to the arithmetic when that distinction is ignored.

2.3 Retention and Private Telemetry

Retention modelling in games is a mature area with a structural dependency. The standard framing takes per-player session logs, engineers behavioural features over an observation window, and predicts whether a player returns. It works, and it requires the operator’s own database. Churn forecasting enhanced with sentiment drawn from Steam reviews is the closest published relative of this work [1], and even there the outcome is a churn label supplied from inside the product.

The consequence for a small studio is awkward. The technique that would answer its question is demonstrated on data it does not hold, and the data it does hold is demonstrated only against outcomes it already knows. The retention proxy in Section 4.4 closes that hole by taking a behavioural quantity, how much longer a player kept playing, out of a field Steam already publishes.

2.4 Community Climate

Toxicity in multiplayer communities has moved from a moderation concern to a measured economic one. The evidence that it costs engagement is now strong and, importantly, causal rather than correlational: Morrier et al. [13] used an instrumental-variable strategy on proprietary Call of Duty match data and report that exposure to toxicity has statistically significant causal effects on short-term player engagement and on the probability that the exposed player uses similar language in the same match, with the size of the effect varying by whether the toxicity came from an opponent or a teammate and by whether the match was won. Player-side accounts point the same way. Coping strategies are widespread and unevenly effective [3], the experience ranges from mildly irritating to genuinely harmful [10], and the intervention systems built in response have been catalogued and found uneven [19]. A recent systematic review notes that the field still lacks a settled definition of the construct it keeps measuring [9], which is a caution worth carrying into any study that puts a number on it.

Two things follow for a study like this one. Toxicity earns its place in a retention model because it is connected to the outcome, and unpleasantness alone would not have justified it. The stronger point is that the evidence for that connection comes from in-game voice and chat logs held by the publisher. Public review text is a weaker instrument: it captures the tone of a game’s storefront discussion rather than conduct inside a match, and treating one as the other would be a misreading. It is, however, the only instrument a studio without a moderation pipeline can actually use.

Measurement itself is comparatively settled. Transformer classifiers trained on the Jigsaw civil-comments corpora are the working standard for scoring text toxicity [2], with rule-based sentiment scoring built for short informal text as a companion measure [7]. Both were designed for comment text rather than game reviews, and Section 4.5 reports what that mismatch costs.

2.5 Leakage and Temporal Evaluation

The output this study needs is a single interpretable number, which puts it in the composite-indicator tradition rather than the pure prediction one. The known issues there concern weighting, aggregation and how far a composite’s ranking survives a change in its own construction [4]. Equal weighting is the usual default absent a strong prior, and the honest response to that default is to report what a data-driven weighting would have produced instead, which Section 6.5 does.

Evaluation carries a warning the rest of this paper takes seriously. Kapoor and Narayanan [8] document leakage as a recurring cause of overstated machine-learning results across many fields, and the variety that applies here is the one where predictor and outcome are not as separate as the model’s feature list suggests. Because the at-risk class is a minority, area under the precision-recall curve is reported beside the ROC figure throughout [14], and feature attribution uses the tree-explainer formulation of Shapley values [12], which supplies the per-game explanation a deployable score needs.

The gap

Three things are known. Early access is a measurable phenomenon whose outcomes have been described but not forecast. Public review text supports good inference about approval and popularity. Hostile community climate reduces engagement, on evidence drawn from private data.

Three things are not. Nobody has asked whether the public record a game leaves during early access anticipates what happens after it launches. Nobody has separated the instrument measuring the predictor from the instrument measuring the outcome, so published effect sizes in this neighbourhood cannot be read as forecasts. And the games that never launched are absent from every sample, so no existing result speaks to the cohort a studio most needs to avoid joining. Those absences are the research questions below.

3 Research Questions

Table 2 states the three questions together with the thing that separates them: the instrument behind each side of the prediction.

RQ1 and RQ2 differ in exactly one respect, and it is the respect that decides how their answers may be read. RQ1 reads predictor and outcome off the same instrument, so a strong result there is an association. RQ2 does not, so a weaker result there is a forecast. RQ3 asks whether the two combine into something a studio can put on a dashboard and act on.

4 Data and Measures

The study is quantitative, observational and retrospective. It uses only data Valve publishes, under terms that permit an application to retrieve and present Steam data to its users [18], and it involves no human participants.

Table 2: The three research questions, and the instruments behind each one.

Question	Predictor from	Outcome from
RQ1 Which early-access engagement signals carry information about post-launch retention?	playtime counters	playtime counters
RQ2 Does community climate in review text anticipate post-launch retention, and post-launch toxicity, beyond the free store-page percentage?	review text	playtime counters, and later review text
RQ3 Does a composite score work as a risk indicator with thresholds committed in advance?	both components	launch outcome and playtime

Table 3: From the Steam catalogue to the modelling frame.

Stage	Games	Removed
Multiplayer titles enumerated from the store	21,330	
too few reviews to measure anything		14,585
never in early access		3,618
no reviews at all		1,018
released too recently for a window		214
no usable listing date		8
Candidates for collection	1,887	
Review histories actually collected	1,793	
of which launched	1,230	
of which still in early access	563	
Launched, ≥ 200 English EA reviews	630	
and ≥ 50 English post-launch reviews	545	
Survivorship cohort (still in EA, ≥ 200)	424	

4.1 Sampling Frame

The population is multiplayer games on Steam that have been in early access at some point. Table 3 traces the whole path from the store listing down to the games actually modelled. Four different counts sit on that path and it is easy to quote the wrong one, so the frame is fixed here and used everywhere afterwards, including in the power statement.

Review histories were collected for 1,793 of those candidates, giving **3,504,533 raw reviews**. Two strategies were used: a cursor walk over the whole history for 1,410 games, and date-bounded windows for the 383 largest, where a single walk would not terminate within the request budget. A per-game cap bound on 341 games, which is recorded and carried into the analysis rather than ignored, because for those titles a row count understates true review volume. Collection was rate-limited, backed off on server errors and checkpointed after every page, so the run could be killed and resumed without loss. It was killed once and resumed, which is why the corpus snapshot carries two run identifiers under one snapshot id.

4.2 Launch-Boundary Recovery

Splitting a game’s reviews into an early-access period and a post-launch period requires knowing when the game left early access. Steam’s `app_release_date` field is not that date. Across the 1,230 launched games in the corpus, the store’s listed date differs from the date the game actually stopped producing early-access reviews

by more than a week in 74.8% of cases, in both directions, with the extremes running from about 456 days early to about 4,251 days late. Using it would have mislabelled three quarters of the sample. We regard this as a finding in its own right: any study that splits Steam titles on the listed release date is splitting most of them in the wrong place.

The transition was instead recovered by binary search over the review stream, probing the early-access flag at successive timestamps until the boundary was bracketed. The procedure took a median of 14 probes and closed the bracket to a median of 15.6 hours. Cohort membership for every review is then decided by comparing its creation timestamp against that derived boundary. It is never decided by the collector’s own window label, which marks whole-history rows uniformly and would silently empty the post-launch cohort for most launched games.

4.3 Review Processing

Cleaning removed markup and URLs, tokenised Steam’s profanity mask rather than deleting it, dropped non-English reviews, dropped anything left empty after cleaning, and hashed the author identifier so no raw Steam ID leaves the raw layer. What survives is 1,865,234 English reviews across the 1,793 games.

Restricting to English is a real cost and is treated as a covariate rather than a solved problem. English share varies from a few percent to over ninety between games, so the per-game share is carried forward as a feature. The multilingual alternative was rejected for a reason specific to this design: the score compares games against each other, and two differently calibrated toxicity models would put a model artefact inside that comparison.

The profanity mask deserves its own note. Steam replaces profanity with heart glyphs before the API returns the review, so a lexicon sentiment scorer reads the sanitised string as positive. Preserving the mask as an explicit token rather than stripping it turns that measurement problem into a feature, and the redaction count becomes an observable signal of censored hostility. Three reviews demonstrate the effect, scored before and after tokenisation:

```
raw +0.983 -> cleaned -0.542 (1 redaction)
"[censored] suck and u should go kil yourself..."
raw +0.942 -> cleaned -0.542 (1 redaction)
"suck me [censored]"
raw +0.813 -> cleaned +0.813 (0 redactions)
"this game is genuinely lovely and i adore it"
```

The third line is the control: a genuinely positive review carries no mask and does not move. The substituted token itself is not neutral to the lexicon, scoring -0.153 on its own, so this is a defensible choice rather than a fix.

4.4 Post-Review Persistence

Every Steam review carries the author’s playtime at the moment of writing and their lifetime playtime at the moment of collection. Post-Review Persistence is the difference: how much longer that player kept playing after they reviewed. Aggregated per game it gives a behavioural retention measure from public data alone.

The distribution forces two choices. It is violently right-skewed, so the median is used rather than the mean, and the 75th percentile is carried alongside it because the median is often pinned at zero. It is pinned at zero because 21.5% of all rows show a player who

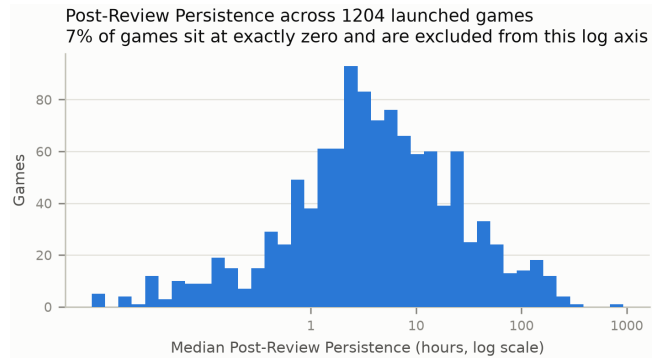


Figure 1: Distribution of Post-Review Persistence across the corpus. The mass at zero is players who never returned after reviewing, and it is the reason the median rather than the mean is used and the reason the zero share is carried as a feature of its own.

never played again after reviewing, which is the churn signal itself and gets its own feature rather than being averaged away. A small residue, 367 rows or 0.0197%, is negative, which is a Steam bookkeeping artefact and is recorded rather than silently clipped. Figure 1 shows the shape that forces these choices.

4.5 Community Climate

Toxicity was scored with Detoxify’s original checkpoint, a BERT model trained on the Jigsaw toxic-comment data [2, 6], and sentiment with the rule-based VADER scorer [7]. Scoring ran in half precision on a single laptop GPU, with reviews length-sorted before batching and reordered afterwards.

A budget of 2,000 reviews per game was imposed. That budget is not a random sample. It is split evenly between the early-access and post-launch cohorts and, within each, allocated across eight equal-width time bins with a floor of 25 reviews per bin, so that a quiet stretch of a game’s life keeps its own representation and the trajectory features stay usable. A uniform random sample would bunch at whichever period was busiest, which destroys those features. The cap bound on 318 games and yielded 1,512,547 scored reviews.

Cleaning was checked against the scores rather than assumed harmless. On a fixed random sample of 200 games, mean toxicity moved from 0.0455 on raw text to 0.0450 on cleaned text, a review-level correlation of 0.9965, while sentiment moved further, from 0.4327 to 0.4028 at a correlation of 0.9367. The larger sentiment shift is the mask correction doing its work.

Throughout, toxicity is treated as a proxy for the climate of a game’s public discussion, not as a measure of in-game conduct, and it is never presented as one.

4.6 Ethics and Privacy

The study uses public secondary data and no human participants. The permission basis is Valve’s own terms of use for the Steam Web API [18]. Author identifiers are hashed at the cleaning stage, which is pseudonymisation rather than anonymisation and is described as such. Raw review text is never redistributed: the released artefact

carries derived, game-level features and the collection code, not the corpus. Toxicity scores describe the climate of a game’s public review discussion, are never presented as a measure of anyone’s conduct, and no per-author score is published. The public-facing surface is read-only and collects nothing from its visitors, which is a design constraint rather than an omission, since a surface that collected input would move the work off the public-secondary-data basis it was approved under.

5 Method

5.1 Outcomes and Temporal Split

The frame is the 545 games in Table 3. The 424 survivorship games are held out entirely and never enter training.

The split is temporal on launch date at seventy percent, giving 381 training and 164 test games, and asserted so that the newest training game is no newer than the oldest test game. A random split or k -fold cross-validation was rejected as the primary design: a studio applies the score to a game that has not launched yet, and random folds let a model learn from the future, which is the deployment condition reversed. Cross-validation is retained as a robustness check only.

Three outcomes are defined by tercile thresholds fitted on training games only and applied unchanged to test: post-launch retention below 2.9 hours of median persistence, post-launch toxicity above 0.0510, and post-launch satisfaction below 0.7416, the last kept only as a robustness check. Because three hypotheses are tested on one dataset, the significance level is Bonferroni-corrected to 0.01667. At $n = 545$ that correction costs almost nothing, moving the detectable correlation at eighty percent power from 0.120 to 0.138.

5.2 Predictors and Leakage Controls

Nineteen predictors enter the models: nine engagement measures, five community measures, three author covariates and two missingness indicators. They are declared as an explicit whitelist in configuration, not derived as “every column except the outcome”. That distinction is deliberate. A blacklist readmits a leaking column the moment a new feature is added, and several columns here leak badly. The reported lifetime review total is a collection-time count containing years of post-launch activity; so are the all-language review count, the collection completeness flag and the cap indicator. All are banned outright.

One confound could not be banned, only measured and declared. Lifetime playtime is a single snapshot taken at collection, so early-access persistence already contains play that happened after launch. The Spearman correlation between early-access and post-launch median persistence is **0.867**. Section 6.2 reports what that does to the results, and it is the reason this paper’s headline is not its highest number.

5.3 Models and Baselines

Models are interpretable linear classifiers. Gradient boosting and a neural alternative were considered and rejected: the deliverable is a score a studio can argue with, at $n = 545$ the extra capacity buys little, and a per-game explanation matters more than a fractional gain.

Two baselines appear throughout. A stratified dummy classifier, returning 0.4977 for retention and 0.4633 for toxicity, establishes the chance floor. The early-access positive-review percentage is the free signal a studio already reads, and it is the number this work has to beat.

Four model families are fitted per outcome where applicable: the baseline alone, the composite health score alone, the full nineteen-feature model, and the cross-instrument model. The cross-instrument model is the design that answers RQ2. It takes five community measures, all derived from review *text*, and predicts post-launch retention, which is measured from a *playtime counter*. No playtime figure enters that model at any point. A sixth variant adds accrual time as a covariate to test the elapsed-time objection.

5.4 Composite Score

The score maps each game’s predictors onto a 0–100 scale, flipping the sign on measures where a higher raw value is worse, and combines an engagement component with a community component. Both weightings are reported: equal weights at 0.50 and 0.50, following the usual default where there is no strong prior [4], and data-driven weights of 0.85 and 0.15 taken from logistic coefficients fitted on training rows only. Scaling is fitted inside the training folds, never over the full dataset. Band cutoffs are set from training terciles and committed before the test set is consulted; cutoffs chosen to maximise test precision would measure the analyst rather than the games.

5.5 Evaluation

The test set was scored once, at the end. Area under the ROC curve is the headline metric, with area under the precision-recall curve reported beside it throughout because the at-risk class is a minority [14], and the Brier score for calibration. Confidence intervals on the cross-instrument gain come from 2,000 bootstrap resamples of the test set. The survivorship comparison uses a one-sided Mann-Whitney test against the Bonferroni-corrected threshold. Feature attribution uses the tree-explainer formulation of Shapley values [12].

5.6 Reproducibility

The pipeline is eight notebooks on Python 3.11.9, one per stage, pinned to a single kernel so the GPU stages cannot silently run under an interpreter without torch. Everything was run once, end to end, on a fixed seed of 42, and every fitted run is logged to a local MLflow store. Every path, threshold, band cutoff and colour lives in one configuration file that the notebooks, the site and the demo application all read, so a value written anywhere else is a defect and the paper, the site and the model cannot disagree about a cutoff. Each notebook ends with an assertion cell checking its own output: row counts, expected columns, unexpected nulls. The modelling stage asserts its own split, frame size and output schema, and fails loudly rather than producing a plausible wrong number.

Each stage reads the previous stage’s file from disk and writes its own; no stage passes a data frame to the next in memory, which is why the interrupted collection run cost nothing. Every destructive file operation is atomic, a discipline adopted after an earlier version

Table 4: All fitted models on the temporal test split. The 0.9008 and 0.9305 figures are associations, not forecasts; this section explains why.

Model	Feat.	Train AUC	Test AUC	AUC- PR	Brier
<i>Retention</i>					
review-% baseline	1	0.6389	0.6019	0.5499	0.2661
health score only	1	0.8381	0.8101	0.7757	0.2135
full features	19	0.9479	0.9008	0.9036	0.1554
cross-instrument	5	0.7291	0.7489	0.7496	0.2226
cross-instr. + accrual	6	0.7311	0.7480	0.7502	0.2278
<i>Community toxicity</i>					
review-% baseline	1	0.6434	0.6486	0.4681	0.2013
health score only	1	0.8217	0.8547	0.7022	0.1433
full features	19	0.9001	0.9305	0.8478	0.0996
<i>Satisfaction (robustness check)</i>					
review-% baseline	1	0.8935	0.9144	0.8959	0.1365
health score only	1	0.7441	0.7054	0.5968	0.2291
full features	19	0.9142	0.9028	0.8879	0.1364

truncated its output file at the start of a run and destroyed a corpus that had taken hours to gather.

The outputs are a game-level feature table of 1,793 rows and 35 columns, a scored table of 969 rows and 36 columns covering the 545 modelled games and the 424 survivorship games, and nineteen result tables, one per reported analysis. Every figure below is generated from those tables rather than drawn by hand, so a regenerated result changes the chart with it. The one exception is the ablation in Figure 2, whose table is retained from an earlier seventeen-predictor run and is no longer rewritten by the modelling stage; it is reported on its own terms and never mixed with the nineteen-predictor figures.

6 Results

All results below are measured on the 164 test games, whose launch dates all fall after every training game’s.

6.1 Descriptive Data

The corpus is 3,504,533 raw reviews over 1,793 games, reduced to 1,865,234 English reviews and 1,512,547 toxicity-scored ones. The modelling frame is 545 launched games with a 381/164 temporal split, plus 424 never-launched games held out entirely.

Two distributional facts shape everything downstream. Post-Review Persistence is pinned at zero for 21.5% of rows (Figure 1), and the at-risk base rate moves between the splits: 33.6% in training against 49.4% in test, because test games are newer and have accrued less playtime against an absolute threshold. Train and test therefore pose measurably different problems, which makes the test figures conservative rather than flattering.

6.2 Shared-Instrument Inflation

Table 4 reports every fitted model. Read down the retention rows first. The health score alone, a single composite number, reaches 0.8101 against the baseline’s 0.6019, and the full nineteen-feature model reaches 0.9008. Toxicity behaves the same way, rising from 0.6486 to 0.9305. Area under the precision-recall curve agrees with the ROC figure on the ordering throughout.

A retention model at 0.9008 invites a claim this paper will not make. Figure 2 shows why. The ablation runs on seventeen of the nineteen predictors: the two missingness indicators are dropped because their attributed importance is exactly zero, leaving nine engagement measures, five community measures and three author covariates. Removing the four persistence features from those seventeen drops retention performance from 0.902 to 0.788, and those four features on their own reach 0.901, which is the entire full model reproduced by four columns. Engagement features alone reach 0.910.

The reason is measurement rather than modelling. Lifetime playtime is a single snapshot taken on the collection date, so early-access persistence necessarily contains play that happened after launch. Early-access and post-launch median persistence correlate at a Spearman ρ of 0.867. The predictor and the outcome are, to a substantial degree, the same quantity read off the same instrument on two disjoint sets of reviewers. This is the variety of leakage Kapoor and Narayanan [8] describe as arising when predictor and outcome are not as separate as the feature list implies, and it is why the 0.9008 and 0.9305 figures are labelled associations throughout and never described as predictive performance.

The mirror image appears in the toxicity column of the same figure, where the persistence family manages 0.673 while community features alone reach 0.933, because early-access and post-launch toxicity share their own instrument. Each outcome is inflated by whichever feature family was measured the same way it was. That symmetry is the evidence that this is a property of the measurement rather than of any one feature set.

6.3 Cross-Instrument Prediction

The way out of that circularity is to force the predictor and the outcome onto different instruments, which is what the cross-instrument model does.

It reaches a test AUC of 0.7489 against the baseline’s 0.6019, a gain of +0.1482 **with a 95% confidence interval of** [+0.0559, +0.2428] over 2,000 bootstrap resamples, and the probability that the gain exceeds zero is 0.998. Figure 3 shows the comparison.

Three objections were put to it and none survives. It is not shared measurement, because text goes in and a playtime counter comes out. It is not elapsed time: training games have a median of 2,350 days for playtime to accumulate against the test games’ 508, but accrual time on its own scores exactly 0.5000, it correlates with the outcome at only 0.143, and adding it as a covariate leaves the result at 0.7480. And it is not the free baseline in disguise, because putting the baseline back into the model gives 0.7428, slightly lower, which says the community signal already contains whatever the store-page percentage knew.

6.4 Calibration and Decision Use

Calibration is honest but imperfect. Figure 4 plots predicted risk against observed rate, and the curve sits above the diagonal across its range: the model is systematically under-confident, understating risk for the games it is least worried about. Its Brier score is 0.2226 against 0.25 for an uninformative forecast, so the probabilities carry real information but should be read as an ordering rather than

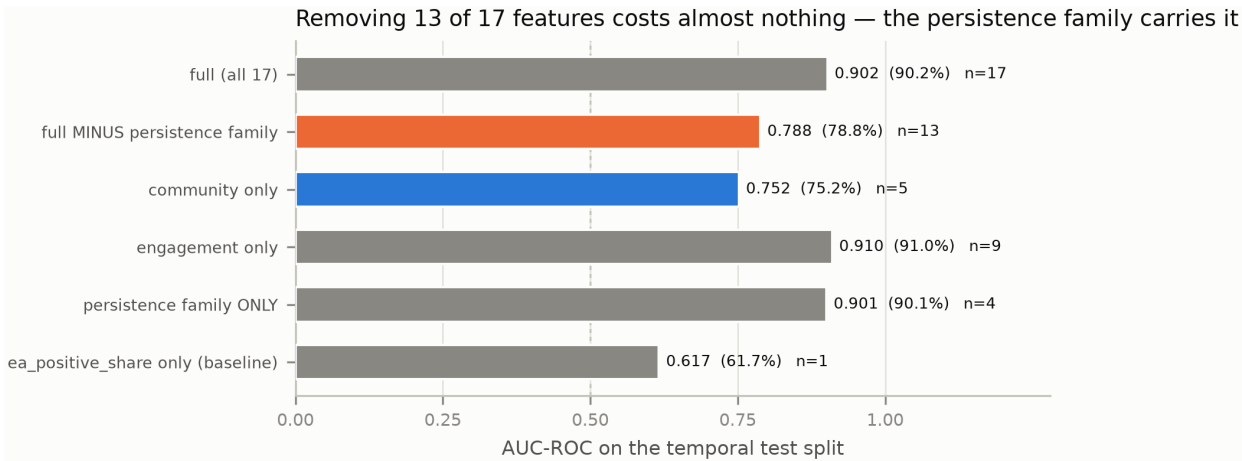


Figure 2: Feature-family ablation on the temporal test split. Four persistence features reproduce the whole retention model, which is a property of the measurement rather than of the model. Run on the seventeen substantive predictors: the two missingness indicators are excluded here because their attributed importance is exactly zero, so the figures differ marginally from Table 4.

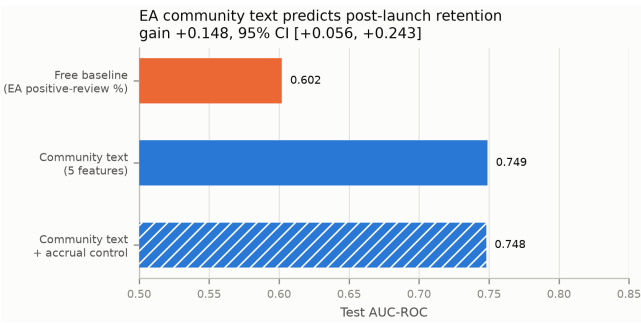


Figure 3: Early-access community text predicts post-launch playtime retention, gaining 0.148 AUC over the free store-page baseline. Predictor and outcome share no measurement instrument, which is what makes this a forecast rather than an association.

as calibrated risks. For a studio deciding where to look first, the ordering is the useful part.

6.5 Score Bands

Weighting was reported both ways. Equal weights across the engagement and community components give scores from 13.8 to 86.6; weights taken from training coefficients give 0.85 to engagement against 0.15 to community and a range of 12.1 to 86.5. The two rankings correlate at 0.8585, which is high enough for the choice not to drive the conclusions and low enough that reporting only one would have been a selective account.

Band cutoffs were fixed from training terciles at 43.6 and 57.2, splitting the training games into 184 at risk, 168 needing attention and 193 healthy, and were committed before the test set was consulted. On test, the At Risk band flags 57 of 164 games at a precision of **0.754** and a recall of 0.531, against a base rate of 0.494. A studio

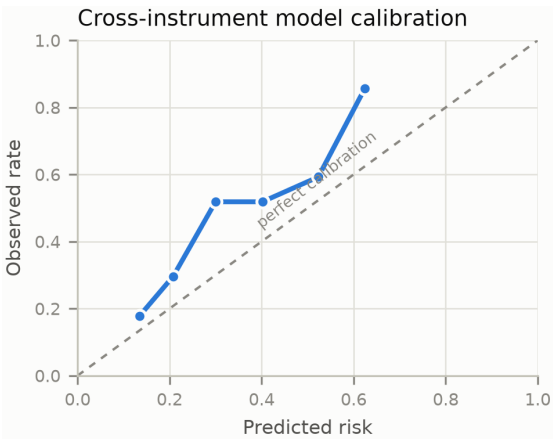


Figure 4: Calibration of the cross-instrument model on the test split. Observed risk exceeds predicted risk throughout, so the model understates rather than overstates danger.

acting on the flag is therefore right about three times in four, where a coin would be right about half the time.

Figure 5 shows the full trade-off. A studio wanting near-certainty can flag only games below 20 and be right every time, catching about a quarter of those truly at risk; one wanting coverage can flag below 85 and catch about 95% at a precision of 0.740. The pre-committed cutoff sits in the middle by construction rather than by hindsight.

Attribution puts the persistence measures at the top, with the 75th percentile of post-review persistence and its median leading, followed by the trend in positive-review share and the share of reviewers who never played again. No size or volume feature appears in the top five, which is the check that a banned column or a proxy for one has not re-entered by the back door.

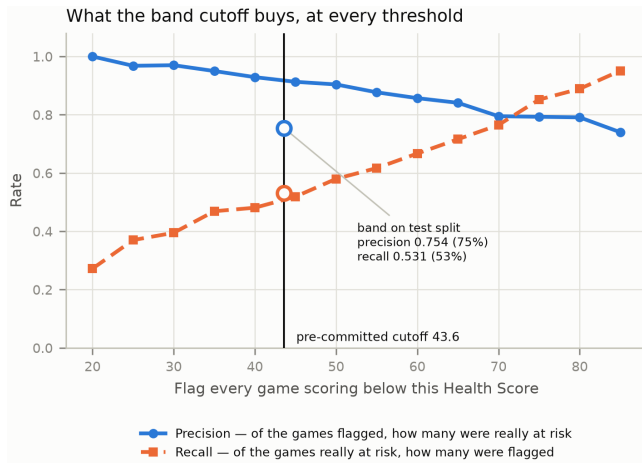


Figure 5: Precision and recall at every possible band cutoff, with the pre-committed cutoff marked. The curve, not the single operating point, is what a studio should be choosing from.

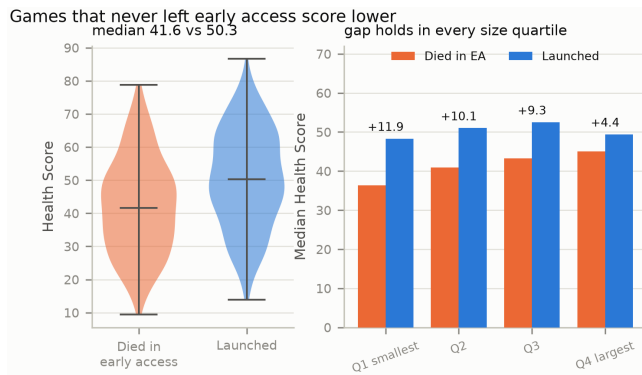


Figure 6: Games that never left early access score lower than games that launched, and the gap survives in every size quartile. The cutoffs were fitted without ever seeing this cohort.

6.6 Never-Launched Cohort

Every result above is measured on games that survived to launch, which is exactly the sample a studio worried about not launching should distrust. The 424 games still in early access with at least 200 early-access reviews were therefore scored on the same fitted ranking and compared against the 545 that launched.

They score lower: a median of 41.6 against 50.3, a gap of 8.8 points, with a one-sided Mann-Whitney test giving $p = 3.3 \times 10^{-16}$, comfortably past the Bonferroni-corrected threshold. The gap is not an artefact of size, since it holds in all four size quartiles at 11.9, 10.1, 9.3 and 4.4 points, each significant on its own. Figure 6 shows both views.

The strength of this test is that the band cutoffs never saw this cohort. They were derived from training terciles among launched games alone, so the abandoned games are being sorted by a rule fitted without them and still land on the correct side of it.

7 Discussion

7.1 Principal Finding

The finding that matters is not the largest number in Table 4, it is the smallest of the ones worth trusting. What players write during early access carries information about whether they will still be playing after launch, and it carries information the store-page percentage does not have. A gain of 0.148 AUC over the free baseline is not a spectacular effect, but it is a real one on an outcome measured with a different instrument, on games the model had never seen, at a confidence interval that excludes zero.

Placed against the literature, that fills a specific hole. Toxicity has been shown to reduce engagement causally on proprietary match telemetry [13], and churn has been forecast from review sentiment against labels supplied from inside the product [1]. This work reaches a comparable conclusion using nothing that is not public, which changes who can act on it. The contribution is not a better model, it is a model a studio without a data team can run.

7.2 Methodological Contribution

The leakage ablation is, on reflection, the most transferable part of the work. Four features reproduced an entire seventeen-feature model, not because the model was good but because the predictor and the outcome were the same measurement twice. The symmetric result on toxicity, where a different feature family inflates a different outcome, shows this is a property of how the quantities are read rather than of any particular feature set.

Published results reporting high AUCs from Steam playtime features without separating instruments are open to the same reading. The diagnostic is cheap: correlate a predictor against its outcome before celebrating the model built on it. Here that would have been one line, and it would have caught the problem before the model was fitted rather than after. The general form of the prescription is to state, for every claimed prediction, which instrument measures the predictor and which measures the outcome, and to treat a shared instrument as disqualifying until shown otherwise.

The launch-boundary result is a smaller contribution of the same kind. Steam's listed release date disagrees with the observed early-access transition for 74.8% of launched titles, so the conventional split point is wrong for three games in four, and any cohort analysis built on it inherits that error.

7.3 Implications for Studios

The practical output is an ordering, not a probability. A studio can compute the score from data it already has access to, read which component drove the band it landed in, and decide where to spend attention. Two caveats bound that claim. The score orders games better than it calibrates them, so it says where to look and not how likely disaster is. And the survivorship comparison is still a comparison of scores, not a demonstration that acting on the score would have changed any outcome. That would need an intervention study, which public data cannot provide.

The case the work is built around is a game with engaged players and a hostile review community. Reporting the engagement and community components separately, rather than only their sum, is

what makes that case visible; a studio reading only its store page would see neither half of the split.

7.4 Implications for Researchers

Three practices transfer. Recover the early-access boundary rather than trusting the listed release date. Declare predictors as a whitelist, because a blacklist readmits a leaking column the moment a feature is added, and four columns in this dataset are collection-time totals containing years of post-launch activity. And analyse the never-launched cohort separately instead of disclosing survivorship bias in a limitations section and moving on: those games are absent from every published sample and they are the ones the question is about.

7.5 Negative Result

The composite adds nothing over the free baseline at predicting satisfaction. The store-page percentage scores 0.9144, the health score alone 0.7054, and the full model 0.9028, below the baseline, and the result is negative at every stage. It is not a surprising result once stated plainly: the positive-review percentage is a direct measure of satisfaction, so asking a nineteen-feature model to beat it there is asking it to beat a thermometer at measuring temperature.

It is reported because it bounds the claim. The score is not a general-purpose quality index, it is a retention and climate instrument, and the satisfaction row is the evidence for that restriction. The useful reading is that these are separable questions. A studio can be well liked and still be losing its population, and the store page will not distinguish the two cases. That is precisely the blind spot the score is for.

8 Limitations

The confound already described is the largest. Persistence is right-censored at the collection date and the censoring window differs by game age, so it is not an early-access-only measure. That is declared, measured at $\rho = 0.867$, and routed around by the cross-instrument model rather than patched.

The base rate shifts between the splits, from 33.6% at risk in training to 49.4% in test, because test games are newer with less accrued playtime and the threshold is absolute. Train and test pose measurably different problems, which makes the test figures conservative.

Restricting to English removes a large share of most games' reviews, and the share varies from a few percent to over ninety. It is carried as a per-game covariate, but a game whose community argues mainly in Russian is being characterised on a minority of its own text. Scoring is also capped at 2,000 reviews per game, binding on 318 of them, so the largest games are measured on a sample where the smallest are measured in full.

Review toxicity measures the climate of a game's storefront discussion, not conduct inside a match. The causal evidence that toxicity costs engagement comes from in-game voice and chat logs [13]; the instrument used here is weaker, and the two must not be read as equivalent. The construct itself also lacks a settled definition in the literature [9].

Two smaller matters are recorded for completeness. The quantile scaler maps any test value below the training minimum to zero,

flattening predictors that trend with calendar time. And the split assertion uses a non-strict inequality, so two distinct games sharing a launch timestamp could sit on opposite sides of the boundary; different games, so not leakage, but worth stating.

Finally, generalisation is bounded by the sample. These are multi-player early-access titles on one storefront with at least 200 English early-access reviews. Nothing here speaks to single-player games, to smaller titles below that threshold, or to any platform other than Steam. The study is observational throughout, so every finding is an association between measured quantities and no causal language is applied to it.

9 Conclusion

The question was whether the public record a game leaves during early access is enough, on its own, to build a legible health score anticipating post-launch retention and community climate. It is, with a qualification that turned out to matter more than the headline.

On RQ1, the signals that best track post-launch retention are the persistence measures. They dominate the attribution and reproduce the whole model on their own, and that last property is the qualification: because lifetime playtime is a snapshot taken at collection, those measures already contain post-launch play, so their apparent strength is largely shared measurement rather than forecasting. Naming that, and measuring it at a rank correlation of 0.867, is a more useful contribution than the 0.9008 it inflates.

On RQ2 the answer is a clean yes. Community climate measured from early-access review text predicts post-launch retention measured from a playtime counter at an AUC of 0.7489 against a free baseline of 0.6019, a gain of 0.148 with a confidence interval excluding zero, and it survives controls for accrual time, game size and the baseline itself. Because the predictor is text and the outcome is a counter, it is a forecast rather than an association, and it is the principal result.

On RQ3 the composite works within stated limits. At a cutoff committed before the test set was seen the At Risk band is right about three times in four against a base rate of one in two, and the 424 games that never left early access score 8.8 points lower than those that launched. It orders games well and calibrates them only roughly, so it belongs in a studio's workflow as a place to look first rather than as a probability to act on. It adds nothing to predicting satisfaction, which the store page measures directly, and that boundary is part of the finding.

Three directions follow. Repeating the collection sweep some months later would replace the censored playtime snapshot with a genuine observation window and settle how much of the persistence signal is real. Extending beyond English, with per-language calibration handled explicitly rather than by exclusion, would remove the largest sampling restriction. And the cheap diagnostic this work stumbled into, correlating each predictor against its outcome before trusting any model built on it, is worth applying to the wider set of published Steam-derived results that report high accuracy without separating their instruments.

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